RESEARCH METHODS PROTOTYPE

VoroGaze AOI

From facial landmarks to defensible areas of interest

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Make the geometry visible before it becomes a number.

Inspectable inputs · deterministic boundaries · traceable outputs

THE PROBLEM

AOI metrics inherit every spatial decision upstream

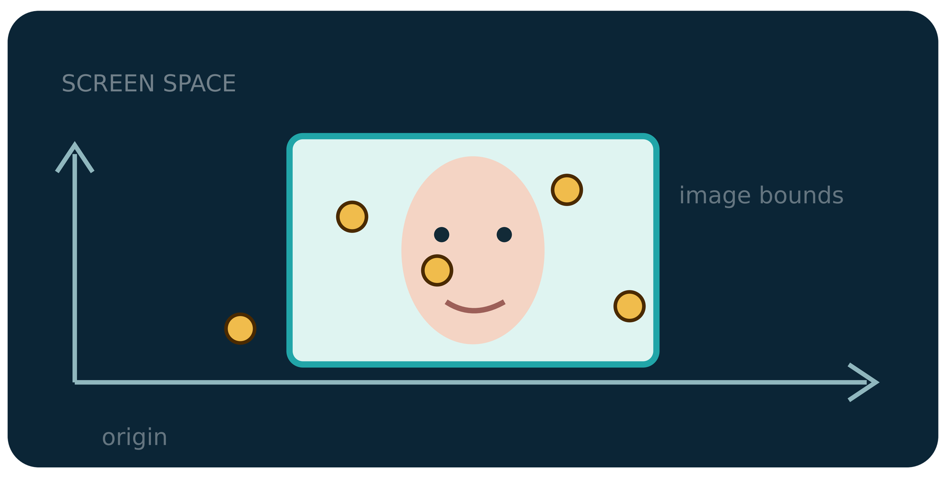
A neat summary can conceal a broken mapping. Before counting fixations, researchers must know that screen, stimulus and gaze occupy one coherent coordinate system.

1. **Which coordinate origin?**
2. **Where was the face placed?**
3. **Which boundary assigned this point?**

VALIDITY FIRST

Geometry before statistics

VoroGaze makes the screen, face placement and fixations visible together – so mismatches can be caught before AOI metrics are computed.



EARLY USABILITY CHECK · N = 2

Useful – with signposting to refine

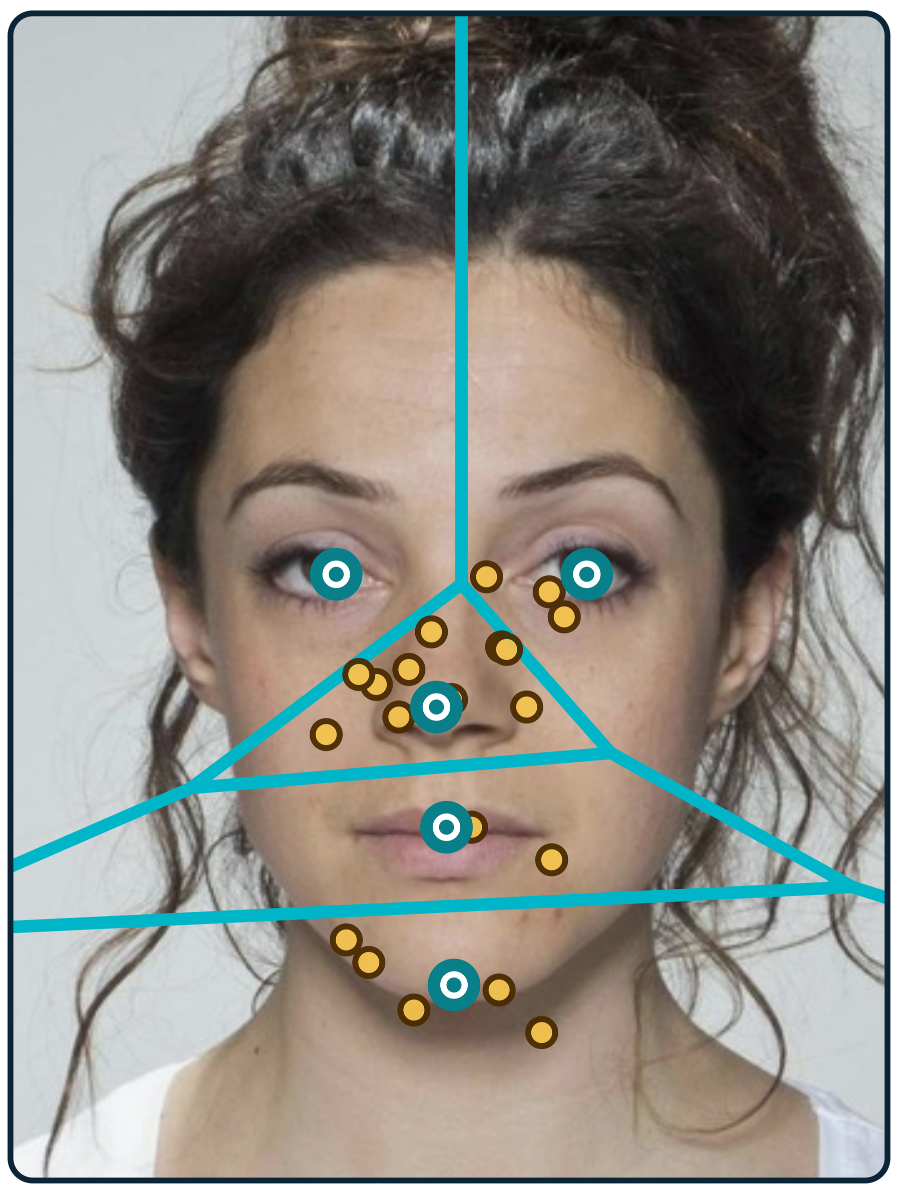
One eye-tracking expert and one non-specialist completed core upload/download tasks. The expert described the tool as useful and self-explanatory.

The non-specialist found AOI assignment after exploration; navigation and output signposting remain explicit design priorities.

WORKED DEMONSTRATION

Landmarks → Voronoi AOIs

The bundled DeBruine et al face keeps every step visible.



landmark centre · fixation · boundary

DEBRUINE ET AL FACE

20 fixations · 2 sessions · 1 face · 5 landmarks

DESCRIPTIVE OUTPUT

The full audit trail precedes the summary

AOI	N fix	total duration	mean duration
left eye	0	—	—
right eye	3	499 ms	166.3
nose	10	1817 ms	181.7
mouth	2	486 ms	243.0
chin	5	1235 ms	247.0
total	**20**	**4037 ms**	—

Worked fixture output – illustrative, not an empirical finding.

A VISIBLE ANALYSIS CHAIN

Inspect → define assign → export

1. **Inspect the inputs**
   Map columns; view screen, face and fixations.
2. **Define meaningful centres**
   Click, label and edit landmarks on the face.
3. **Assign transparently**
   Nearest centre = the point's Voronoi cell.
4. **Export the audit trail**
   Retain row-level assignments, then summarise.

Geometry first · metrics second

WHAT BECOMES INSPECTABLE

Deterministic after the choices

- ✓ Saved centres + image bounds reproduce the same bounded partition.
- ✓ Every fixation retains its source row and assigned AOI before aggregation.
- ✓ Validity still depends on defensible landmark choices, preprocessing and study design.

PROTOTYPE – METHODS WORKFLOW, NOT EMPIRICAL VALIDATION

TRY VOROGAZE

Choose your route online

QR opens the guide:
screencast · worked example
Research Workbench

Workbench: vorogaze-workbench.mjgreen.uk
Worked example: vorogaze-example.mjgreen.uk
Guide + screencast: vorogaze.mjgreen.uk

TAKE-HOME MESSAGE

VoroGaze turns AOI construction into an inspectable analysis chain – make the geometry visible, make the boundaries reproducible, then compute the metrics.

Repository: github.com/mjgreen/vorogazeaoi · Worked example uses the bundled DeBruine et al face only.